

Madhav Khanal

 madhav2k22 |  madhav-khanal |  khanalmadhav323@gmail.com |  +977-9867726632

SUMMARY

AI/ML Developer and Data Scientist with experience in machine learning projects. Skilled in Python, C, Java, FastAPI, and TensorFlow/Keras. Passionate about building AI-powered applications and delivering production-ready solutions.

WORK EXPERIENCE

Data Science Intern (Training Program) — Fusemachines Nepal 2024

Performed data cleaning, preprocessing, and exploratory data analysis (EDA) on real-world datasets using Python. Built and evaluated machine learning models using statistical and supervised learning techniques, applied feature engineering, and visualized insights to support data-driven decision making. Collaborated in an industry-oriented environment to understand business problems and translate them into data science solutions.

PROJECTS

Exploratory Data Analysis on Real Datasets [Link to Demo](#)

Executed end-to-end EDA on the Titanic and Flight datasets using Python (Pandas, Seaborn, Matplotlib). Performed univariate and bivariate analysis to identify outliers and missing patterns, and utilized multi-variate techniques like correlation matrices to detect interaction effects

Trinetra – Automated Security Vulnerability Detection [Link to Demo](#)

An AI-powered platform developed to identify and remediate security vulnerabilities in C programs. The system features a full-stack architecture using React and an asynchronous FastAPI backend with secure authentication and admin management. Performed functional, API, and UI testing; designed test cases, reported bugs, and conducted regression and performance testing to ensure system reliability and accuracy..

FakeNews Detection [Link to Demo](#)

Built an end-to-end NLP Fake News Detector using Python. Implemented text preprocessing (tokenization, stop-word removal) and TF-IDF vectorization to train classification models. Successfully identified deceptive linguistic patterns, achieving high model accuracy

EDUCATION

2021 - 2025	Bachelor of Software Engineering Degree at Nepal College of Information Technology (GPA: 3.57/4.0)	
2020	Class 12th NEB	(3.47)
2017	Class 10th NEB	(3.60)

SKILLS

Programming Languages	C, C++, Python, Java
Web Technologies	FastAPI, REST APIs, TailwindCSS, HTML, CSS
AI / Machine Learning	CNNs, Transformers, Deep Learning, Model Evaluation, Data Preprocessing
Frameworks & Libraries	TensorFlow, Keras, PyTorch, SQLAlchemy, Pydantic